Forensics – Assignment 2

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For this Forensic Examination, I will answer the outlined questions, while detailing the steps I took to answer each, including the initial setup of my working environment.

I will do this by including the volatility command, the output of the command, a description of the approach taken with the chosen plugin, and a summary of my understanding & justification regarding my findings.

Setup

First, I created a Virtual Machine using Oracle VirtualBox, downloading an image of Ubuntu Desktop (24.04.03) from the following link. Next, I setup the VM in line with the outlined omissions:

- ***Guest Additions***
- ***Folder Sharing***
- ***Cut & Paste***
- ***Internet Connectivity***
- ***Any Sharing Features***

Here are screenshots of proof of omission following the setup (including the download of files & required applications such as Volatility & the Memory Image):

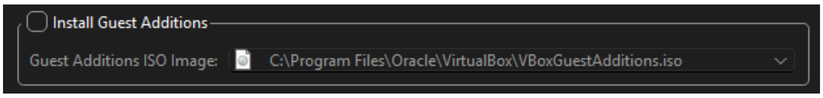


Figure 1 - Guest Additions

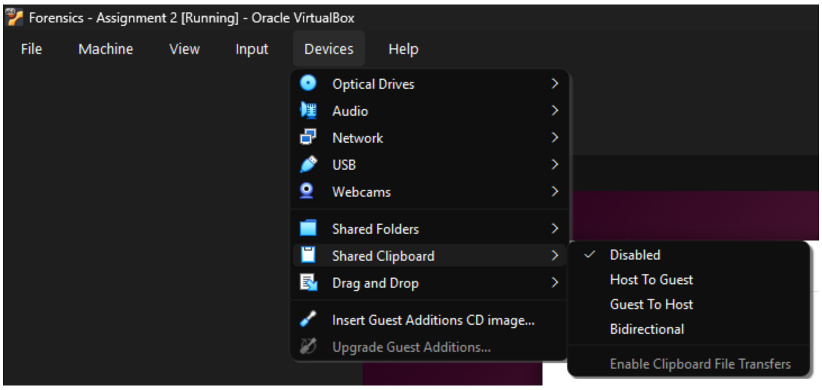


Figure 2 - Cut & Paste

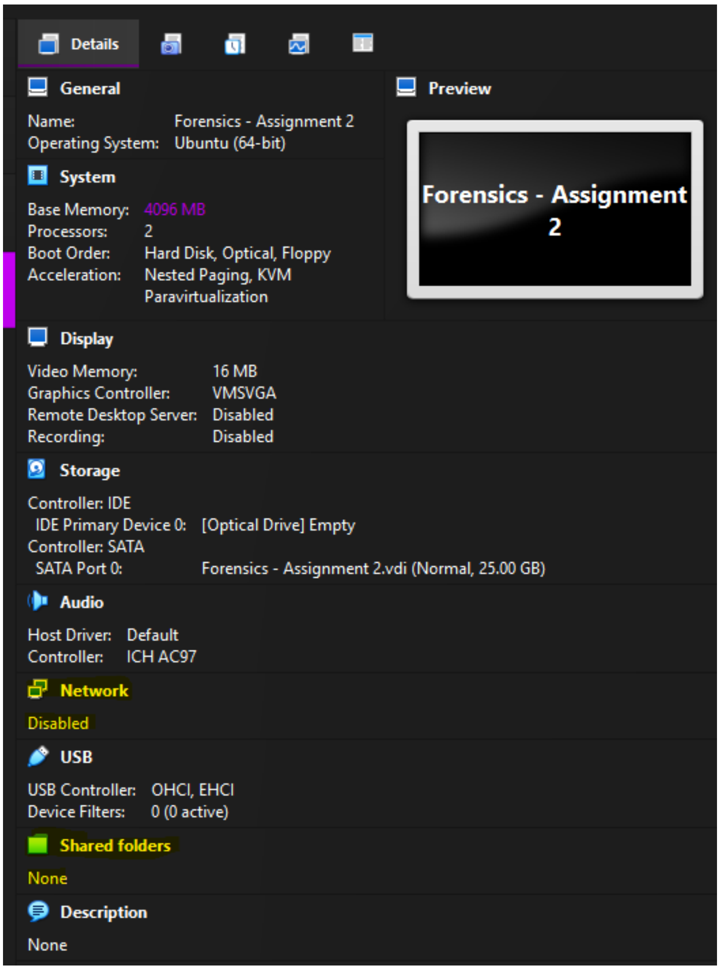


Figure 3 - Folder Sharing & Internet Connectivity

After filling out the Virtual Machine installation details (such as username devuser & password devpass), I started the Ubuntu installation process. Once the installation process was completed, a restart was performed and I logged into the virtual machine.

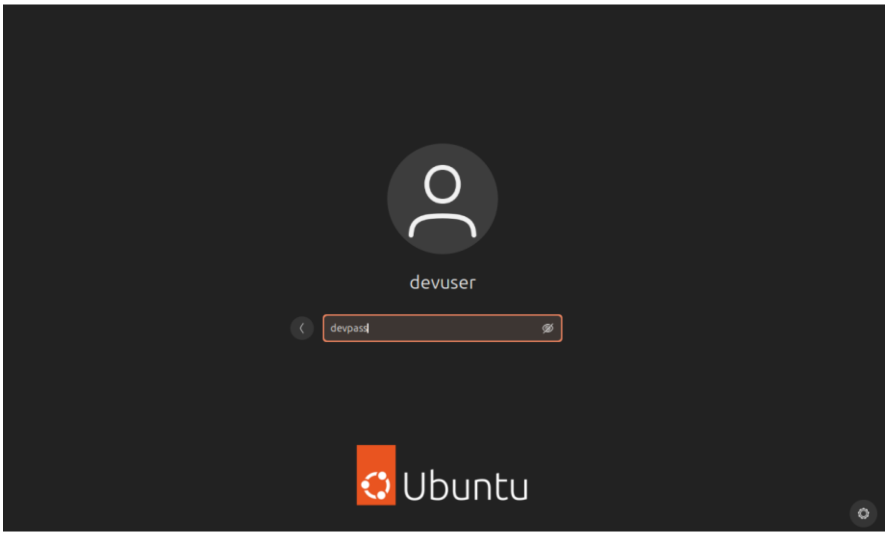


Figure 4 - Login Screen

From here, I downloaded Volatility (2.6) through their GitHub repository (github.com/volatilityfoundation/volatility/releases) with the volatility_2.6_lin64_standalone.zip file.

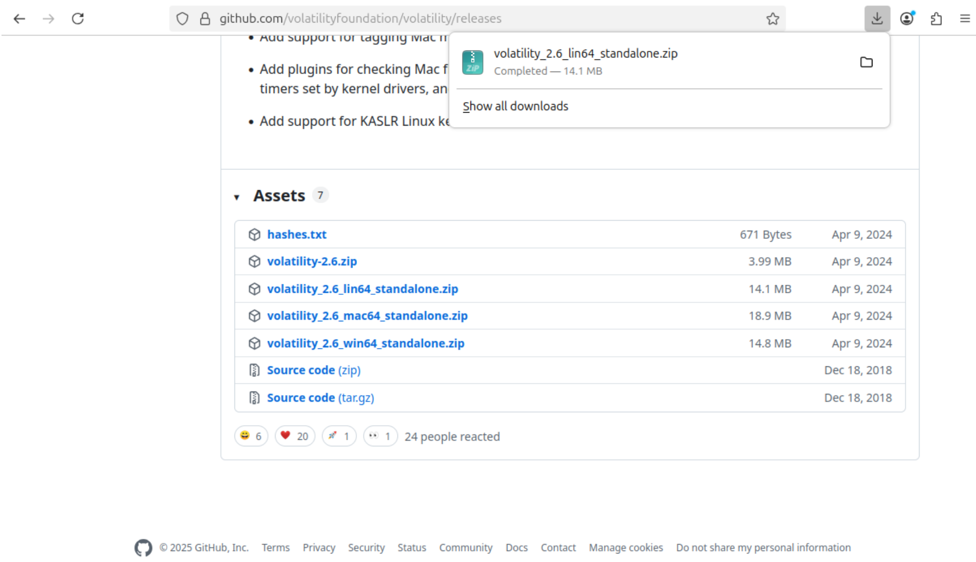


Figure 5 - Volatility ZIP Download

I extracted the file and renamed some paths to make it easier to navigate in the command line interface (Home/AssignmentII/Vol/workspace/volMain)

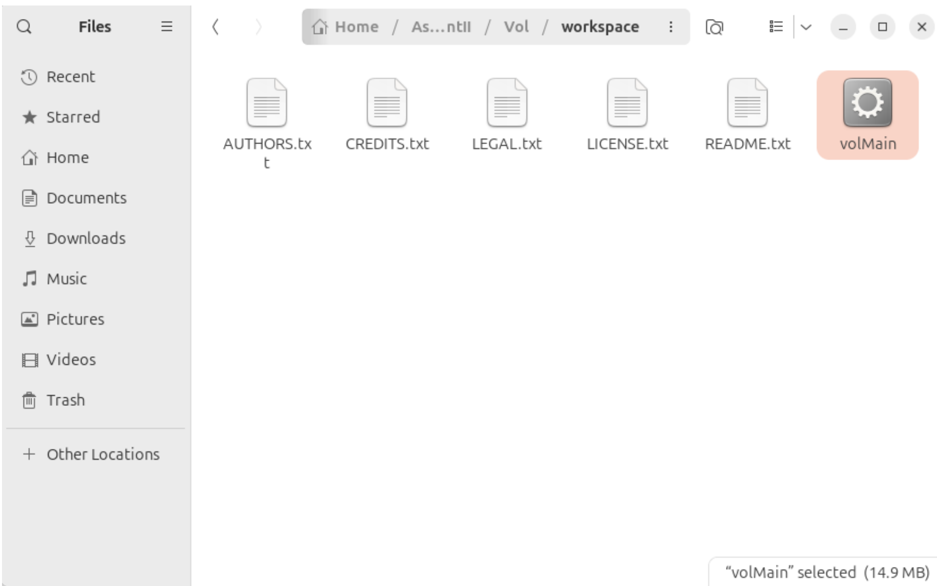


Figure 6 - Volatility Workspace

Once that was completed, I then obtained the required RAM image through the given NIST link (https://cfreds-archive.nist.gov/mem/memory-images.rar).

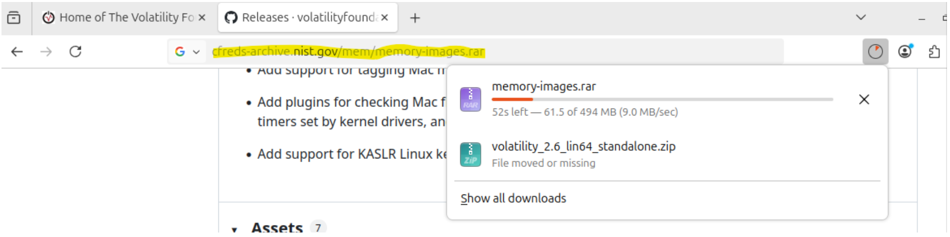


Figure 7 - Memory Images Download

Once that, was downloaded, I moved it into my workspace folder and downloaded WinRAR to extract the images, to obtain the one I would be working with: xp-laptop-2005-06-25.img

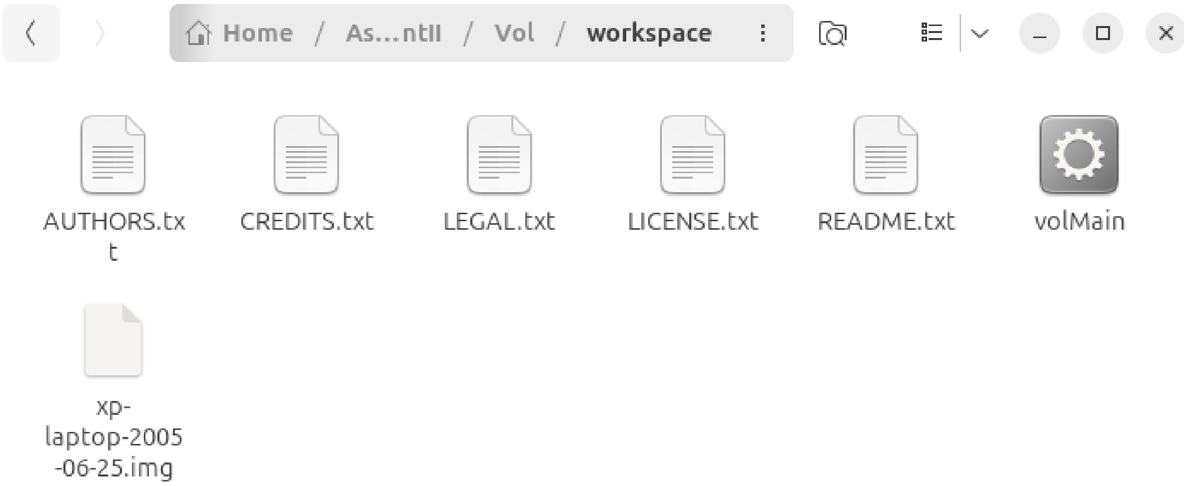
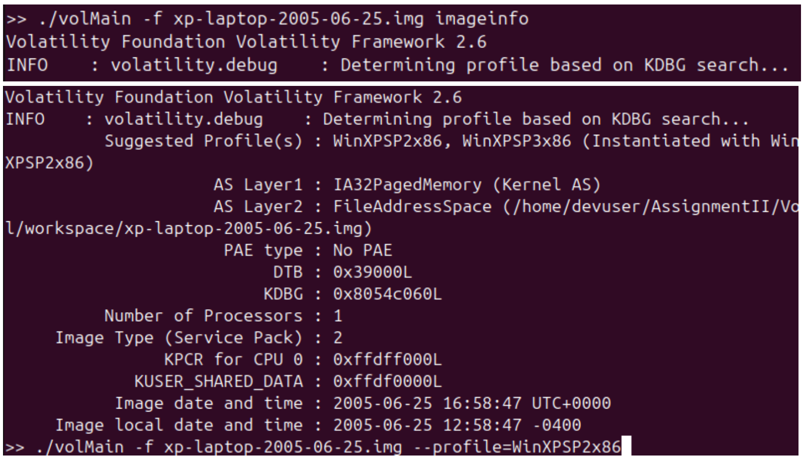


Figure 8 - Extracted Memory Image

With the setup process now completed, with all files gathered and programs installed, I closed the virtual machine and disabled the network to remove Internet access for examination purposes as asked, completing the outlined exclusion requirements.

On the next page, I will begin answering the posed questions.



Questions

Question 1

The initial top three processes I would prioritize for deeper analysis would be the following:

- ***dd.exe***
- ***cmd.exe***
- ***wmiprvse.exe***

I have chosen these three processes because:

- **dd.exe** was the last executable ran, and it was run from the command line (found with parent process id 2624). It is not a typical Windows Application and being run from the command line raises suspicion. Assuming it is like the dd command found with UNIX (and Linux etc.) operating systems, it would be the clearest example of an executable (aside from the listed Firefox.exe) that does not come with the operating system, with a function that raises potential suspicion depending on usage.
- While **cmd.exe** is not in itself suspicious, given that it was used to run dd.exe means that it should be checked out to see what was going on at the time. The history of the usage should be examined.
- **wmiprvse.exe** was run around the same time as cmd.exe and dd.exe. While, like cmd.exe, it is a typical part of Windows functionality, it could also be abused. With the suspicious dd.exe executable being run soon after, and the nature of this process, providing the ability to check / edit environment variables, view / terminate processes, obtain the MAC Address & Serial Number of the computer, it could have been prepared for misuse. It is also one of the only listed processes without a session ID.

While these are the chosen three, I would point out the **Fast.exe** executable as well, which was typically used to switch user accounts without logging in or out between accounts. There is potential for a bad actor to have accessed the machine remotely, or maybe the person stood away from their desk, and a bad actor used dd.exe (and assuming it is like UNIX dd) through the command line to edit or copy files on the machine. There are several potential possibilities with this close combination of processes.

Other potential options include the **Pluck** processes (PluckTray.exe, PluckUpdater.ex, PluckSvr.exe) which I cannot find much information, with only this link, which I cannot even be sure is reliable.

While the Pluck processes are unknown, that doesn't necessarily mean they are dangerous, but the lack of information around them means I cannot be certain of their nature. They could very well be the **three** worthy of deeper analysis, but I can't really tell from looking at just that file.

I used the following command to pull the data into a file:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 pslist > commands
/pslist.txt
```

Which produced the following output in a .txt file (snippet):

```
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x823c87c0 System                    4      0     61     1140 ------      0
0x81fdf020 smss.exe                448      4      3       21 ------      0 2005-06-25 16:47:28 UTC+0000
0x81f5a3b8 csrss.exe               504    448     12      596      0      0 2005-06-25 16:47:30 UTC+0000
0x81f8eb10 winlogon.exe            528    448     21      508      0      0 2005-06-25 16:47:31 UTC+0000
0x820e0da0 services.exe            580    528     18      401      0      0 2005-06-25 16:47:31 UTC+0000
0x82199668 lsass.exe               592    528     21      374      0      0 2005-06-25 16:47:31 UTC+0000
0x81fa5aa0 svchost.exe             740    580     17      198      0      0 2005-06-25 16:47:32 UTC+0000
0x81fa8650 svchost.exe             800    580     10      302      0      0 2005-06-25 16:47:33 UTC+0000
0x81faba78 svchost.exe             840    580     83     1589      0      0 2005-06-25 16:47:33 UTC+0000
0x81fa8240 Smc.exe                 876    580     22      423      0      0 2005-06-25 16:47:33 UTC+0000
0x81f8dda0 svchost.exe             984    580      6       90      0      0 2005-06-25 16:47:35 UTC+0000
0x81f6e7e8 svchost.exe            1024    580     15      207      0      0 2005-06-25 16:47:35 UTC+0000
0x81f9a670 spoolsv.exe            1224    580     12      136      0      0 2005-06-25 16:47:39 UTC+0000
0x81f5f020 ssonsvr.exe            1632   1580      1       24      0      0 2005-06-25 16:47:46 UTC+0000
0x8202bda0 explorer.exe           1812   1764     22      553      0      0 2005-06-25 16:47:47 UTC+0000
0x82113c48 Directcd.exe           1936   1812      4       40      0      0 2005-06-25 16:47:48 UTC+0000
0x81f67500 TaskSwitch.exe         1952   1812      1       21      0      0 2005-06-25 16:47:48 UTC+0000
0x81f6ca90 Fast.exe               1960   1812      1       22      0      0 2005-06-25 16:47:48 UTC+0000
0x820dd588 VPTray.exe             1980   1812      2       89      0      0 2005-06-25 16:47:49 UTC+0000
0x82025608 atiptaxx.exe           2040   1812      1       51      0      0 2005-06-25 16:47:49 UTC+0000
0x81faf280 jusched.exe             188   1812      1       22      0      0 2005-06-25 16:47:49 UTC+0000
0x821125d0 EM_EXEC.EXE             224    112      2       74      0      0 2005-06-25 16:47:50 UTC+0000
0x82076558 ati2evxx.exe            432    580      4       38      0      0 2005-06-25 16:47:55 UTC+0000
0x81f68518 Crypserv.exe            688    580      3       34      0      0 2005-06-25 16:47:55 UTC+0000
0x82059da0 DefWatch.exe            864    580      3       27      0      0 2005-06-25 16:47:55 UTC+0000
0x81f6db28 msdtc.exe              1076    580     14      166      0      0 2005-06-25 16:47:55 UTC+0000
0x82021a78 Rtvscan.exe            1304    580     37      300      0      0 2005-06-25 16:47:58 UTC+0000
0x81f48da0 tcpsvcs.exe            1400    580      2       94      0      0 2005-06-25 16:47:58 UTC+0000
0x820238e0 snmp.exe               1424    580      5      192      0      0 2005-06-25 16:47:58 UTC+0000
0x82081da0 svchost.exe            1484    580      6      119      0      0 2005-06-25 16:47:59 UTC+0000
0x821ca3d0 wdfmgr.exe             1548    580      4       65      0      0 2005-06-25 16:47:59 UTC+0000
0x821ce4d8 Fast.exe               1700    580      2       32      0      0 2005-06-25 16:48:01 UTC+0000
0x821d4da0 mqsvc.exe              1948    580     23      205      0      0 2005-06-25 16:48:02 UTC+0000
0x81343790 mqtgsvc.exe            2536    580      9      119      0      0 2005-06-25 16:48:05 UTC+0000
0xffab8020 alg.exe                2868    580      6      108      0      0 2005-06-25 16:48:11 UTC+0000
0x8205eda0 wuauclt.exe            2424    840      4      160      0      0 2005-06-25 16:49:21 UTC+0000
0xffaa0c10 firefox.exe            2160   1812      6      182      0      0 2005-06-25 16:49:22 UTC+0000
0x82218020 PluckSvr.exe            944    740      9      227      0      0 2005-06-25 16:51:00 UTC+0000
0x814b13b0 iexplore.exe           2392   1812      9      365      0      0 2005-06-25 16:51:02 UTC+0000
0x81ed76b0 PluckTray.exe          2740    944      3      105      0      0 2005-06-25 16:51:10 UTC+0000
0x81f269e0 PluckUpdater.ex        3076   1812      0 --------      0      0 2005-06-25 16:51:15 UTC+0000   2005-06-25 16:51:30 UTC+0000
0xffadc9d0 PluckUpdater.ex        1916    944      0 --------      0      0 2005-06-25 16:51:40 UTC+0000   2005-06-25 16:53:49 UTC+0000
0x821fb3b8 PluckTray.exe          3256   1812      0 --------      0      0 2005-06-25 16:54:28 UTC+0000   2005-06-25 16:54:28 UTC+0000
0x82079c18 cmd.exe                2624   1812      1       29      0      0 2005-06-25 16:57:36 UTC+0000
0x82000980 wmiprvse.exe           4080    740      7        0 ------      0 2005-06-25 16:57:53 UTC+0000
0x822148f0 PluckTray.exe          3100   1812      0 --------      0      0 2005-06-25 16:57:59 UTC+0000   2005-06-25 16:57:59 UTC+0000
0x81ed84e8 dd.exe                 4012   2624      1       22      0      0 2005-06-25 16:58:46 UTC+0000
```

Question 2

The processes terminated in memory with associated timestamps are the following:

- **PluckUpdater.ex** - 2 Each - 25/06/2005 at 16:51:30 & 16:53:49 respectively
- **PluckTray.exe** - 2 Each – 25/06/2005 at 16:54:28 & 16:57:59 respectively

To find this out, I used the **psscan** command, to find all process objects left in memory, including exited processes, which could be left out by pslist:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 psscan > commands/psscan.txt
Volatility Foundation Volatility Framework 2.6
>>
```

The following is portions of the output, containing all the exited processes, as shown in the 'Time exited' column:

Offset(P)	Name	PID	PPID	PDB	Time created	Time exited
0x0000000001343790	mqtgsvc.exe	2536	580	0x17406000	2005-06-25 16:48:05 UTC+0000	
0x00000000014b13b0	iexplore.exe	2392	1812	0x16f8f000	2005-06-25 16:51:02 UTC+0000	
0x0000000001ed76b0	PluckTray.exe	2740	944	0x175fc000	2005-06-25 16:51:10 UTC+0000	
0x0000000001ed84e8	dd.exe	4012	2624	0x0eee8000	2005-06-25 16:58:46 UTC+0000	
0x0000000001f269e0	PluckUpdater.ex	3076	1812	0x1a6c5000	2005-06-25 16:51:15 UTC+0000	2005-06-25 16:51:30 UTC+0000
0x00000000021ce4d8	Fast.exe	1700	580	0x15c85000	2005-06-25 16:48:01 UTC+0000	
0x00000000021d4da0	mqsvc.exe	1948	580	0x164e2000	2005-06-25 16:48:02 UTC+0000	
0x00000000021fb3b8	PluckTray.exe	3256	1812	0x17c92000	2005-06-25 16:54:28 UTC+0000	2005-06-25 16:54:28 UTC+0000
0x00000000022148f0	PluckTray.exe	3100	1812	0x1119f000	2005-06-25 16:57:59 UTC+0000	2005-06-25 16:57:59 UTC+0000
0x0000000002218020	PluckSvr.exe	944	740	0x14a78000	2005-06-25 16:51:00 UTC+0000	
0x00000000023c87c0	System	4	0	0x00039000		
0x0000000004096da0	svchost.exe	1484	580	0x1515f000	2005-06-25 16:47:59 UTC+0000	
0x000000000ee763b0	iexplore.exe	2392	1812	0x16f8f000	2005-06-25 16:51:02 UTC+0000	
0x000000000f55d670	spoolsv.exe	1224	580	0x1147b000	2005-06-25 16:47:39 UTC+0000	
0x000000000fe5f8e0	snmp.exe	1424	580	0x14f3a000	2005-06-25 16:47:58 UTC+0000	
0x0000000012cd3020	smss.exe	448	4	0x0c55a000	2005-06-25 16:47:28 UTC+0000	
0x0000000012d67a90	Fast.exe	1960	1812	0x13aaf000	2005-06-25 16:47:48 UTC+0000	
0x00000000131f0da0	svchost.exe	984	580	0x10220000	2005-06-25 16:47:35 UTC+0000	
0x0000000013a36a78	svchost.exe	840	580	0x0ea71000	2005-06-25 16:47:33 UTC+0000	
0x0000000013a597e8	svchost.exe	1024	580	0x1043e000	2005-06-25 16:47:35 UTC+0000	
0x0000000013f924e8	dd.exe	4012	2624	0x0eee8000	2005-06-25 16:58:46 UTC+0000	
0x0000000016c7f9d0	PluckUpdater.ex	1916	944	0x1ba0e000	2005-06-25 16:51:40 UTC+0000	2005-06-25 16:53:49 UTC+0000
0x00000000171033b0	iexplore.exe	2392	1812	0x16f8f000	2005-06-25 16:51:02 UTC+0000	
0x0000000017fdb020	alg.exe	2868	580	0x18679000	2005-06-25 16:48:11 UTC+0000	

The other processes listed (without times exited at) were processes which were not exited before the capture of this memory image.

Having briefly mentioned the Pluck processes in the previous section, it is curious that they are the contained terminated processes.

I also examined with psxview:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 psxview
```

```
Volatility Foundation Volatility Framework 2.6
Offset(P)  Name                    PID pslist psscan thrdproc pspcid csrss session deskthrd ExitTime
---------- -------------------- ------ ------ ------ -------- ------ ----- ------- -------- --------
0x01f67500 TaskSwitch.exe         1952 True   True   True     True   True  True    True
0x01faf280 jusched.exe             188 True   True   True     True   True  True    True
0x021ca3d0 wdfmgr.exe             1548 True   True   True     True   True  True    True
0x02081da0 svchost.exe            1484 True   True   True     True   True  True    True
0x020dd588 VPTray.exe             1980 True   True   True     True   True  True    True
0x17fdb020 alg.exe                2868 True   True   True     True   True  True    True
0x01f8eb10 winlogon.exe            528 True   True   True     True   True  True    True
0x02079c18 cmd.exe                2624 True   True   True     True   True  True    True
0x01f68518 Crypserv.exe            688 True   True   True     True   True  True    True
0x01fa5aa0 svchost.exe             740 True   True   True     True   True  True    True
0x020e0da0 services.exe            580 True   True   True     True   True  True    True
0x014b13b0 iexplore.exe           2392 True   True   True     True   True  True    True
0x01343790 mqtgsvc.exe            2536 True   True   True     True   True  True    True
0x01f48da0 tcpsvcs.exe            1400 True   True   True     True   True  True    True
0x01f6db28 msdtc.exe              1076 True   True   True     True   True  True    False
0x01ed76b0 PluckTray.exe          2740 True   True   True     True   True  True    True
0x02025608 atiptaxx.exe           2040 True   True   True     True   True  True    True
0x0202bda0 explorer.exe           1812 True   True   True     True   True  True    True
0x01f8dda0 svchost.exe             984 True   True   True     True   True  True    False
0x01f6ca90 Fast.exe               1960 True   True   True     True   True  True    True
0x01fa8240 Smc.exe                 876 True   True   True     True   True  True    True
0x01f5f020 ssonsvr.exe            1632 True   True   True     True   True  True    True
0x186fec10 firefox.exe            2160 True   True   True     True   True  True    True
0x02218020 PluckSvr.exe            944 True   True   True     True   True  True    True
0x02113c48 Directcd.exe           1936 True   True   True     True   True  True    True
0x01fa8650 svchost.exe             800 True   True   True     True   True  True    False
0x02021a78 Rtvscan.exe            1304 True   True   True     True   True  True    True
0x021d4da0 mqsvc.exe              1948 True   True   True     True   True  True    True
0x02076558 ati2evxx.exe            432 True   True   True     True   True  True    True
0x01ed84e8 dd.exe                 4012 True   True   True     True   True  True    True
0x020238e0 snmp.exe               1424 True   True   True     True   True  True    True
0x021125d0 EM_EXEC.EXE             224 True   True   True     True   True  True    True
0x02059da0 DefWatch.exe            864 True   True   True     True   True  True    True
0x02199668 lsass.exe               592 True   True   True     True   True  True    True
0x01f9a670 spoolsv.exe            1224 True   True   True     True   True  True    True
0x01f6e7e8 svchost.exe            1024 True   True   True     True   True  True    True
0x01faba78 svchost.exe             840 True   True   True     True   True  True    True
0x0205eda0 wuauclt.exe            2424 True   True   True     True   True  True    True
0x021ce4d8 Fast.exe               1700 True   True   True     True   True  True    True
0x01f269e0 PluckUpdater.ex        3076 True   True   False    True   False False   False    2005-06-25 16:51:30 UTC+0000
0x16c7f9d0 PluckUpdater.ex        1916 True   True   False    True   False False   False    2005-06-25 16:53:49 UTC+0000
0x01f5a3b8 csrss.exe               504 True   True   True     True   False True    True
0x023c87c0 System                    4 True   True   True     True   False False   False
```

```
0x021fb3b8 PluckTray.exe          3256 True   True   False    True   False False   False    2005-06-25 16:54:28 UTC+0000
0x022148f0 PluckTray.exe          3100 True   True   False    True   False False   False    2005-06-25 16:57:59 UTC+0000
0x02000980 wmiprvse.exe           4080 True   True   True     False  False True    True
0x12cd3020 smss.exe                448 False  True   False    False  False False   False
0x0fe5f8e0 snmp.exe               1424 False  True   False    False  False False   False
0x131f0da0 svchost.exe             984 False  True   False    False  False False   False
0x18899da0 svchost.exe             984 False  True   False    False  False False   False
0x1b4db020 smss.exe                448 False  True   False    False  False False   False
0x12d67a90 Fast.exe               1960 False  True   False    False  False False   False
0x0ee763b0 iexplore.exe           2392 False  True   False    False  False False   False
0x13a36a78 svchost.exe             840 False  True   False    False  False False   False
0x1a192a90 Fast.exe               1960 False  True   False    False  False False   False
0x0f55d670 spoolsv.exe            1224 False  True   False    False  False False   False
0x1e5b2670 spoolsv.exe            1224 False  True   False    False  False False   False
0x04096da0 svchost.exe            1484 False  True   False    False  False False   False
0x171033b0 iexplore.exe           2392 False  True   False    False  False False   False
0x13f924e8 dd.exe                 4012 False  True   False    False  False False   False
0x13a597e8 svchost.exe            1024 False  True   False    False  False False   False
```

The second screenshot, displaying processes **False on pslist** but **True for psscan**, appear to be **terminated** or **dead** processes. None of these have **associated timestamps** however, starting with **smss.exe** and ending with **svchost.exe**.

Question 3

Here is the list of all **DLLs** loaded by **dd.exe**:

```
************************************************************************
dd.exe pid:   4012
Command line : dd if=\\.\PhysicalMemory of=c:\xp-laptop-2005-06-25.img conv=noerror
Service Pack 2

Base             Size  LoadCount Path
---------- ---------- ---------- ----
0x00400000     0xe000     0xffff D:\dd\UnicodeRelease\dd.exe
0x7c900000    0xb0000     0xffff C:\WINDOWS\system32\ntdll.dll
0x7c800000    0xf4000     0xffff C:\WINDOWS\system32\kernel32.dll
0x10000000     0x6000     0xffff D:\dd\UnicodeRelease\getopt.dll
0x7c000000    0x54000     0xffff D:\dd\UnicodeRelease\MSVCR70.dll
0x77c00000     0x8000     0xffff C:\WINDOWS\system32\VERSION.dll
0x77fe0000    0x11000     0xffff C:\WINDOWS\system32\Secur32.dll
0x77dd0000    0x9b000     0xffff C:\WINDOWS\system32\ADVAPI32.dll
0x77e70000    0x91000     0xffff C:\WINDOWS\system32\RPCRT4.dll
0x774e0000   0x13d000     0xffff C:\WINDOWS\system32\ole32.dll
0x77f10000    0x46000     0xffff C:\WINDOWS\system32\GDI32.dll
0x77d40000    0x90000     0xffff C:\WINDOWS\system32\USER32.dll
0x77c10000    0x58000     0xffff C:\WINDOWS\system32\msvcrt.dll
```

Looking at the shown dll files, found inside the dd folder and system32 folder, there doesn't appear to be any potentially malicious (suspicious) files. Each file appears to be a standard Windows dll file or a file that could be associated with a valid dd Windows version.

The suspicion that could arise is if these files were copied or altered in some way, as files within system32 are critical to Windows operating systems, but their appearance here doesn't mean that is the case, as I don't know the exact nature of the dd.exe and how it works.

I found these DLLs using the **dllist** command shown below:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 dlllist -p 4012 > commands/dllist.txt
Volatility Foundation Volatility Framework 2.6
>>
```

Question 4

The network activity patterns associated with PID 1916 (PluckUpdater.exe) are:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connscan
Volatility Foundation Volatility Framework 2.6
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x01370e70 192.168.2.7:1115          207.126.123.29:80         1916
0x01ed1a50 3.0.48.2:17985            66.179.81.245:20084       4287933200
0x01f0e358 192.168.2.7:1164          66.179.81.247:80          944
0x01f11e70 192.168.2.7:1082          205.161.7.134:80          2392
0x01f35cd0 192.168.2.7:1086          199.239.137.200:80        1916
0x01f88e70 192.168.2.7:1162          170.224.8.51:80           1916
0x020869b0 127.0.0.1:1055            127.0.0.1:1056            2160
0x021ca8b8 192.168.2.7:1116          66.161.12.81:80           1916
0x021d2e70 192.168.2.7:1161          66.135.211.87:443         1916
0x02201800 192.168.2.7:1091          209.73.26.183:80          1916
0x02207ab0 192.168.2.7:1151          66.150.96.111:80          1916
0x0220c008 192.168.2.7:1077          64.62.243.144:80          2392
0x0220d6b8 192.168.2.7:1066          199.239.137.200:80        2392
0x02210c48 192.168.2.7:1157          66.151.149.10:80          1916
0x02889800 192.168.2.7:1091          209.73.26.183:80          1916
0x108d2e70 192.168.2.7:1115          207.126.123.29:80         1916
0x187a8008 192.168.2.7:1155          66.35.250.150:80          1916
0x18fffaf0 127.0.0.1:1056            127.0.0.1:1055            2160
0x1d5bde70 192.168.2.7:1115          207.126.123.29:80         1916
0x1f4eb008 192.168.2.7:1155          66.35.250.150:80          1916
```

This output was found by the command in the above and below screenshots:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connscan
```

The following commands were also used to check network activity but provided no potential links to the given process ID:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 sockets
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connections
```

These two above commands were used to check for listening sockets and active TCP connections.

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 sockscan
```

This above command can be used to find residual data regarding sockets, which also provided me with no relevant data.

Now, with the **connscan** command, thirteen associations were made to the 1916 Process ID. These appear to be multiple outbound connections made by the PluckUpdater process from the same address through a variety of ports.

Question 5

There appears to be no evidence of a connection to **192.168.1.100:443**. Using the same commands as before (connections & connscan), there does not appear to be a usage of this address and HTTPS port for connection.

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connections
Volatility Foundation Volatility Framework 2.6
Offset(V)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x820869b0 127.0.0.1:1055            127.0.0.1:1056            2160
0xffa2baf0 127.0.0.1:1056            127.0.0.1:1055            2160
0x8220c008 192.168.2.7:1077          64.62.243.144:80          2392
0x81f11e70 192.168.2.7:1082          205.161.7.134:80          2392
0x8220d6b8 192.168.2.7:1066          199.239.137.200:80        2392
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connscan
Volatility Foundation Volatility Framework 2.6
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x01370e70 192.168.2.7:1115          207.126.123.29:80         1916
0x01ed1a50 3.0.48.2:17985            66.179.81.245:20084       4287933200
0x01f0e358 192.168.2.7:1164          66.179.81.247:80          944
0x01f11e70 192.168.2.7:1082          205.161.7.134:80          2392
0x01f35cd0 192.168.2.7:1086          199.239.137.200:80        1916
0x01f88e70 192.168.2.7:1162          170.224.8.51:80           1916
0x020869b0 127.0.0.1:1055            127.0.0.1:1056            2160
0x021ca8b8 192.168.2.7:1116          66.161.12.81:80           1916
0x021d2e70 192.168.2.7:1161          66.135.211.87:443         1916
0x02201800 192.168.2.7:1091          209.73.26.183:80          1916
0x02207ab0 192.168.2.7:1151          66.150.96.111:80          1916
0x0220c008 192.168.2.7:1077          64.62.243.144:80          2392
0x0220d6b8 192.168.2.7:1066          199.239.137.200:80        2392
0x02210c48 192.168.2.7:1157          66.151.149.10:80          1916
0x02889800 192.168.2.7:1091          209.73.26.183:80          1916
0x108d2e70 192.168.2.7:1115          207.126.123.29:80         1916
0x187a8008 192.168.2.7:1155          66.35.250.150:80          1916
0x18fffaf0 127.0.0.1:1056            127.0.0.1:1055            2160
0x1d5bde70 192.168.2.7:1115          207.126.123.29:80         1916
0x1f4eb008 192.168.2.7:1155          66.35.250.150:80          1916
```

The only seemingly **associated HTTPS** port usage **(:443)** is linked to **PID 1916 (PluckUpdater.exe)** with a connection from **192.168.2.7:1161** to the remote address of **66.135.211.87:443**.

Question 6

After inspecting **NTUSER.DATA** to find what URLs were manually typed into the Internet Explorer address bar, I could only find one entry, which will be shown in the **last screenshot** of the question.

Initially, I used the following command to pull the **registry hives** for this memory capture:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 hivelist
Volatility Foundation Volatility Framework 2.6
Virtual    Physical   Name
---------- ---------- ----
0xe1ecd008 0x11221008 \Device\HarddiskVolume1\Documents and Settings\Sarah\Local Settings\Application Data
\Microsoft\Windows\UsrClass.dat
0xe1eff758 0x1294a758 \Device\HarddiskVolume1\Documents and Settings\Sarah\NTUSER.DAT
0xe1bf9008 0x0e6d0008 \Device\HarddiskVolume1\Documents and Settings\LocalService\Local Settings\Applicati
on Data\Microsoft\Windows\UsrClass.dat
0xe1c26850 0x0e882850 \Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT
0xe1bf1b60 0x0e213b60 \Device\HarddiskVolume1\Documents and Settings\NetworkService\Local Settings\Applica
tion Data\Microsoft\Windows\UsrClass.dat
0xe1c2a758 0x0e88e758 \Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT
0xe1982008 0x0c61d008 \Device\HarddiskVolume1\WINDOWS\system32\config\software
0xe197f758 0x0c622758 \Device\HarddiskVolume1\WINDOWS\system32\config\default
0xe1986008 0x0c632008 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM
0xe197a758 0x0c60e758 \Device\HarddiskVolume1\WINDOWS\system32\config\SECURITY
0xe1558578 0x02d63578 [no name]
0xe1035b60 0x0283db60 \Device\HarddiskVolume1\WINDOWS\system32\config\system
0xe102e008 0x02837008 [no name]
```

The current user appears to be **Sarah**, so I examined further using **printkey**:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 printkey -K "Software\Microsoft\Internet Explorer"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable   (V) = Volatile
```

Here, I found the necessary directory: **TypedURLs**

```
Registry: \Device\HarddiskVolume1\Documents and Settings\Sarah\NTUSER.DAT
Key name: Internet Explorer (S)
Last updated: 2005-06-25 16:51:12 UTC+0000

Subkeys:
  (S) Desktop
  (S) Document Windows
  (S) Download
  (S) Explorer Bars
  (S) Extensions
  (S) Help_Menu_URLs
  (S) International
  (S) Main
  (S) New Windows
  (S) SearchUrl
  (S) Security
  (S) Services
  (S) Settings
  (S) Toolbar
  (S) TypedURLs
  (S) URLSearchHooks
```

Looking inside, I discovered the following:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 printkey -K "Software\Microsoft\Internet Explorer\TypedURLs"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \Device\HarddiskVolume1\Documents and Settings\Sarah\NTUSER.DAT
Key name: TypedURLs (S)
Last updated: 2005-06-05 21:07:09 UTC+0000

Subkeys:

Values:
REG_SZ        url1            : (S) http://www.microsoft.com/isapi/redir.dll?prd=ie&pver=6&ar=msnhome
```

It appears to be some generic redirect page for **MSN**, which is a generic web portal provided by Microsoft.

From this gathered data, there is a possibility that other history was potentially wiped, or I have missed it.

Question 7

To grab the list of all autorun entries from **HKLM…\Run**, I first ran the hivelist command once again, to find the config software directory (the equivalent of **HKLM\Software**).

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 hivelist
Volatility Foundation Volatility Framework 2.6
Virtual    Physical   Name
---------- ---------- ----
0xe1ecd008 0x11221008 \Device\HarddiskVolume1\Documents and Settings\Sarah\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat
0xe1eff758 0x1294a758 \Device\HarddiskVolume1\Documents and Settings\Sarah\NTUSER.DAT
0xe1bf9008 0x0e6d0008 \Device\HarddiskVolume1\Documents and Settings\LocalService\Local Settings\Application Data\Microsoft\Windows\UsrClass.
0xe1c26850 0x0e882850 \Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT
0xe1bf1b60 0x0e213b60 \Device\HarddiskVolume1\Documents and Settings\NetworkService\Local Settings\Application Data\Microsoft\Windows\UsrClas
0xe1c2a758 0x0e88e758 \Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT
0xe1982008 0x0c61d008 \Device\HarddiskVolume1\WINDOWS\system32\config\software
0xe197f758 0x0c622758 \Device\HarddiskVolume1\WINDOWS\system32\config\default
0xe1986008 0x0c632008 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM
0xe197a758 0x0c60e758 \Device\HarddiskVolume1\WINDOWS\system32\config\SECURITY
0xe1558578 0x02d63578 [no name]
0xe1035b60 0x0283db60 \Device\HarddiskVolume1\WINDOWS\system32\config\system
0xe102e008 0x02837008 [no name]
```

Having gotten this, I used the following command to access the autorun entries from Run:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 printkey -o 0xe1982008 -K "Microsoft\Windows\CurrentVersion\Run"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \Device\HarddiskVolume1\WINDOWS\system32\config\software
Key name: Run (S)
Last updated: 2005-06-05 21:08:16 UTC+0000

Subkeys:
  (S) OptionalComponents

Values:
REG_SZ       MsmqIntCert     : (S) regsvr32 /s mqrt.dll
REG_SZ       AdaptecDirectCD : (S) "C:\Program Files\Roxio\Easy CD Creator 5\DirectCD\DirectCD.exe"
REG_SZ       BackgroundSwitcher : (S) C:\WINDOWS\System32\bgswitch.exe
REG_SZ       CoolSwitch      : (S) C:\WINDOWS\System32\taskswitch.exe
REG_SZ       FastUser        : (S) C:\WINDOWS\System32\fast.exe
REG_SZ       Pictometry      : (S) C:\WINDOWS\pictometry.vbs
REG_SZ       vptray          : (S) C:\PROGRA~1\SYMANT~1\SYMANT~1\vptray.exe
REG_SZ       QuickTime Task  : (S) "C:\Program Files\QuickTime\qttask.exe" -atboottime
REG_SZ       SmcService      : (S) C:\PROGRA~1\Sygate\SPF\smc.exe -startgui
REG_SZ       ISLP2STA.EXE    : (S) ISLP2STA.EXE START
REG_SZ       AtiPTA          : (S) Atiptaxx.exe
REG_SZ       Logitech Utility : (S) Logi_MwX.Exe
REG_SZ       SunJavaUpdateSched : (S) C:\Program Files\Java\jre1.5.0_02\bin\jusched.exe
REG_SZ       TkBellExe       : (S) "C:\Program Files\Common Files\Real\Update_OB\realsched.exe"  -osboot
```

From this, the autorun entries are shown:

- **MsmqIntCert**
- **AdaptecDirectCD**
- **BackgroundSwitcher**
- **CoolSwitch**
- **FastUser**
- **Pictometry**
- **Vptray**
- **QuickTime Task**
- **SmcService**
- **ISLP2STA.EXE**
- **AtiPTA**
- **Logitech Utility**
- **SunJavaUpdateSched**
- **TkBellExe**

Question 8

Using the '**filescan**' plugin, here is a list of all **GIF** files found in memory:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 filescan | grep -i ".gif"
Volatility Foundation Volatility Framework 2.6
0x000000000133f5b8      1      0 R--rwd \Device\HarddiskVolume1[illegible]n Files\Mozilla Firefox\res\loading-image.gif
0x000000000135f028      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\go_button[1].gif
0x0000000001373f90      1      0 -W-rwd \Device\HarddiskVolume1 [illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\ZU3H5UQ0\xml_36x14[1].gif
0x0000000001395be0      1      0 -W-rwd \Device\HarddiskVolume [illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\readers_reviews_sub2[1].gif
0x00000000014b8668      1      0 R--rwd \Device\HarddiskVolume1\Program Files\Mozilla Firefox\searchplugins\google.gif
0x0000000001ed1498      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\line468[1].gif
0x0000000001edb8f0      1      0 -W-rwd \Device\HarddiskVolume1[illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\gn_features[1].gif
0x0000000001ee4390      1      0 -W-rwd \Device\HarddiskVolume1\Documents and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\gn_news[1].gif
0x0000000001f1a528      1      0 -W-rwd \Device\HarddiskVolume1%[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\spacer[1].gif
0x0000000001f1aef8      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\gn_services[1].gif
0x0000000001f346a0      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\ZU3H5UQ0\gn_opinion[1].gif
0x0000000001f35930      1      0 -W-rwd \Device\HarddiskVolume1\Documents and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\mm_09[1].gif
0x0000000002052728      1      0 -W-rwd \Device\HarddiskVolume1[illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\sports[1].gif
0x0000000002118320      1      0 -W-rwd \Device\HarddiskVolume1[illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\26A8QE4D\mm_10[1].gif
0x000000000212d340      1      0 R--rwd \Device\HarddiskVolume1\Program Files\Crash.Net Desktop Race Babe\skin\channelhead.gif
0x000000000212d540      1      0 R--rwd \Device\HarddiskVolume1\Program Files\Crash.Net Desktop Race Babe\skin\channeltail.gif
0x000000000220aa50      1      0 -W-rwd \Device\HarddiskVolume1ff[illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\inside[1].gif
0x000000000220fa50      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\mm_1b[1].gif
0x0000000002216530      1      0 -W-rwd \Device\HarddiskVolume1 [illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\automobiles[1].gif
0x0000000002216f90      1      0 -W-rwd \Device\HarddiskVolume1[illegible]nts and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\jobs[1].gif
0x0000000002259c50      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\mm_15[1].gif
0x000000000e27e390      1      0 -W-rwd \Device\HarddiskVolume1\Documents and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\G8GE3JQQ\gn_news[1].gif
0x000000000eab0028      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\go_button[1].gif
0x000000000f15e6a0      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\ZU3H5UQ0\gn_opinion[1].gif
0x0000000011cce028      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\D33YZADF\go_button[1].gif
0x0000000012575668      1      0 R--rwd \Device\HarddiskVolume1\Program Files\Mozilla Firefox\searchplugins\google.gif
0x00000000188aa668      1      0 R--rwd \Device\HarddiskVolume1\Program Files\Mozilla Firefox\searchplugins\google.gif
0x0000000018d8f6a0      1      0 -W-rwd \Device\HarddiskVolume1[illegible]ments and Settings\Sarah\Local Settings\Temporary Internet Files\Content.IE5\ZU3H5UQ0\gn_opinion[1].gif
>>
```

There is a total of **28** (shown above) different GIF files, found inside Temporary Internet Files (Content.IE5) directories, the Crash.Net Desktop Race Babe skin directory & Mozilla Firefox directories.

After running the following command to identify file handles (which process is handling which files):

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 handles -t File > commands/gif-check.txt
Volatility Foundation Volatility Framework 2.6
```

There were no results found. This was done by initially attempting to filter by .gif:

```
>> grep -i ".gif" commands/gif-check.txt
```

And then manually searching for the extension:

```
Open                              • gif-check.txt
                     ~/AssignmentII/Vol/workspace/commands
0x82083818    592     0x50c   0x1f01ff File        \Device\    .gif
0x821cb610    592     0x514   0x1200a0 File        \Device\    Regular expressions  Case sensitive  Match whole word only
0x82083488    592     0x524   0x12019f File        \Device\
0x81fa3f90    592     0x530   0x1f01ff File        \Device\Afd\Endpoint
0x82082310    592     0x538   0x1f01ff File        \Device\Udp
0x821c2cc8    592     0x564   0x1f01ff File        \Device\Afd\Endpoint
0x821c2ad8    592     0x568   0x1f01ff File        \Device\Udp
0x820eecc0    592     0x578   0x1f01ff File        \Device\Afd\Endpoint
0x820eec28    592     0x57c   0x1f01ff File        \Device\RawIp\255
0x82081d10    592     0x658   0x12019f File        \Device\NamedPipe\lsass
0x81f9c7f8    740       0xc   0x100020 File        \Device\Harddisk-Volume1\WINDOWS\system32
0x81f933d0    740      0x4c   0x100001 File        \Device\KsecDD
0x81f937f8    740      0x64   0x100020 File
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.2180_x-
ww_a84f1ff9
0x81f98a88    740      0xb8   0x12019f File        \Device\NamedPipe\net\NtControlPipe1
0x82109658    740     0x104   0x100000 File        \Device\Dfs
0x81fd44b0    740     0x2a4   0x12019f File        \Device\Termdd
0x81fd3660    740     0x2ac   0x12019f File        \Device\Termdd
0x81fbdc70    740     0x2e8   0x12019f File        \Device\NamedPipe\Ctx_WinStation_API_service
0x81fcc478    740     0x2ec   0x12019f File        \Device\NamedPipe\Ctx_WinStation_API_service
0x821aeb38    740     0x2f8   0x12019f File        \Device\Termdd
0x821ae028    740     0x304   0x12019f File        \Device\Termdd
0x821aa888    800       0xc   0x100020 File        \Device\HarddiskVolume1\WINDOWS\system32
0x823cd4f0    800      0x48   0x100001 File        \Device\KsecDD
0x8239fc38    800      0x5c   0x100020 File
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.2180_x-
```

From this, we can conclude that there were no relevant processes attached to GIF files at the time of this memory capture, unless I missed something.

Question 9

After inspecting the files opened by **dd.exe**, with a process ID of **4012**, using the **handles** command once again, here is the list of files / devices accessed, alongside the evidence available of the process activity:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 handles -p 4012 -t File
Volatility Foundation Volatility Framework 2.6
Offset(V)     Pid     Handle     Access Type             Details
---------- ------ ---------- ---------- ---------------- -------
0x821742e0   4012       0x74   0x100020 File             \Device\DP(1)0-0+b\dd\UnicodeRelease
0x821ae240   4012       0x7ac  0x12019f File             \Device\HarddiskVolume1\xp-laptop-2005-06-25.img
0x81f1b7f8   4012       0x7c8  0x100001 File             \Device\KsecDD
```

The **first line** appears to just be the directory for dd.exe. The **second line** appears to be the image I am currently analysing, while the **last line** is the Kernel-Mode Security Support Provider Interface for Windows.

This appears to show that dd was used to create the forensic memory image that I am currently using.

I also used the following command to check the **console history**, being aware that the program was started in the **command line**:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 consoles
```

Question 10

Here are the IP addresses linked with iexplore.exe, which has a process ID of 2392 (found in previous sections), using **connscan**:

- **205.161.7.134:80**
- **64.62.243.144:80**
- **199.239.137.200:80**

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 connscan | grep "2392"
Volatility Foundation Volatility Framework 2.6
0x01f11e70 192.168.2.7:1082          205.161.7.134:80          2392
0x0220c008 192.168.2.7:1077          64.62.243.144:80          2392
0x0220d6b8 192.168.2.7:1066          199.239.137.200:80        2392
```

That is the command I used, filtering for 2392.

Question 11

To find the usernames of users on the computer, I will pull the offset for the **Security Account Manager**, which is used to store login credentials with Windows, and System, to use the key used for encrypting the hashes in SAM, using the below hivelist command:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 hivelist | egrep "config"
Volatility Foundation Volatility Framework 2.6
0xe1982008 0x0c61d008 \Device\HarddiskVolume1\WINDOWS\system32\config\software
0xe197f758 0x0c622758 \Device\HarddiskVolume1\WINDOWS\system32\config\default
0xe1986008 0x0c632008 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM
0xe197a758 0x0c60e758 \Device\HarddiskVolume1\WINDOWS\system32\config\SECURITY
0xe1035b60 0x0283db60 \Device\HarddiskVolume1\WINDOWS\system32\config\system
```

With this, I will use the **SAM** & **system** offsets with **hashdump**:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 hashdump -y 0xe1035b60 -s 0xe1986008
Volatility Foundation Volatility Framework 2.6
Administrator:500:08f3a52bdd35f179c81667e9d738c5d9:ed88cccbc08d1c18bcded317112555f4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
HelpAssistant:1000:ddd4c9c883a8ecb2078f88d729ba2e67:e78d693bc40f92a534197dc1d3a6d34f:::
SUPPORT_388945a0:1002:aad3b435b51404eeaad3b435b51404ee:8bfd47482583168a0ae5ab020e1186a9:::
phoenix:1003:07b8418e83fad948aad3b435b51404ee:53905140b80b6d8cbe1ab5953f7c1c51:::
ASPNET:1004:2b5f618079400df84f9346ce3e830467:aef73a8bb65a0f01d9470fadc55a411c:::
Sarah:1006:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
>>
```

-y uses the system offset to read the boot key while **-s** uses the readkey to decrypt the password hashes in SAM to display the username (at the beginning of the listed string)

- **Administrator**
- **Guest**
- **HelpAssistant**
- **SUPPORT_388945a0**
- **Phoenix**
- **ASPNET**
- **Sarah**

Question 12

To get the volume of D Drive & the serial number, I used the following command, which I used earlier, previously spotting the information relevant to this question:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 consoles
```

The volume in the D: Drive is **KORNBLUM (492MB+)** and the volume serial number is **D4A8-A3D7**.

The previous commands used by the user left the information in for me.

```
D:\dd\UnicodeRelease>dir
 Volume in drive D is KORNBLUM
 Volume Serial Number is D4A8-A3D7

 Directory of D:\dd\UnicodeRelease

06/25/2005  12:55 PM    <DIR>          .
06/25/2005  12:55 PM    <DIR>          ..
06/25/2005  12:55 PM            57,344 dd.exe
06/25/2005  12:55 PM             9,216 getopt.dll
06/25/2005  12:55 PM            14,848 md5lib.dll
06/25/2005  12:55 PM            17,408 md5sum.exe
06/25/2005  12:55 PM            73,728 nc.exe
06/25/2005  12:55 PM            15,872 volume_dump.exe
06/25/2005  12:55 PM           233,472 wipe.exe
06/25/2005  12:55 PM            51,200 zlibU.dll
06/25/2005  12:55 PM             6,148 .DS_Store
06/25/2005  12:55 PM         2,313,216 MSVCR70.PDB
06/25/2005  12:55 PM           487,424 MSVCP70.DLL
06/25/2005  12:55 PM         2,837,504 MSVCP70.PDB
06/25/2005  12:55 PM           344,064 msvcr70.dll
              13 File(s)      6,461,444 bytes
               2 Dir(s)     492,994,560 bytes free
```

Question 13

The computer hostname is **SPLATITUDE**.

This was found by using the envars command for environment variables, by filtering the Local Security Authority Subsystem Service (lsass.exe, which handles system authentication) process and searching for **COMPUTERNAME**:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 envars -p 592 | grep COMPUTERNAME
Volatility Foundation Volatility Framework 2.6
     592 lsass.exe               0x00010000 COMPUTERNAME                     SPLATITUDE
```

The same name can be found throughout environment variables, repeated multiple times:

```
>> ./volMain -f xp-laptop-2005-06-25.img --profile=WinXPSP2x86 envars
```

```
 1936 Directcd.exe          0x00010000 TMP                           C:\DOCUME~1\Sarah\LOCALS~1\Temp
 1936 Directcd.exe          0x00010000 USERDOMAIN                    SPLATITUDE
 1936 Directcd.exe          0x00010000 USERNAME                      Sarah
 1936 Directcd.exe          0x00010000 USERPROFILE                   C:\Documents and Settings\Sarah
 1936 Directcd.exe          0x00010000 windir                        C:\WINDOWS
 1952 TaskSwitch.exe        0x00010000 ALLUSERSPROFILE               C:\Documents and Settings\All Users
 1952 TaskSwitch.exe        0x00010000 APPDATA                       C:\Documents and Settings\Sarah\Application Data
 1952 TaskSwitch.exe        0x00010000 CLIENTNAME                    Console
 1952 TaskSwitch.exe        0x00010000 CommonProgramFiles            C:\Program Files\Common Files
 1952 TaskSwitch.exe        0x00010000 COMPUTERNAME                  SPLATITUDE
 1952 TaskSwitch.exe        0x00010000 ComSpec                       C:\WINDOWS\system32\cmd.exe
 1952 TaskSwitch.exe        0x00010000 FP_NO_HOST_CHECK              NO
 1952 TaskSwitch.exe        0x00010000 HOMEDRIVE                     C:
 1952 TaskSwitch.exe        0x00010000 HOMEPATH                      \Documents and Settings\Sarah
 1952 TaskSwitch.exe        0x00010000 LOGONSERVER                   \\SPLATITUDE
 1952 TaskSwitch.exe        0x00010000 NUMBER_OF_PROCESSORS          1
 1952 TaskSwitch.exe        0x00010000 OS                            Windows_NT
```

```
 688 Crypserv.exe          0x00010000 COMPUTERNAME                  SPLATITUDE
 688 Crypserv.exe          0x00010000 ComSpec                       C:\WINDOWS\system32\cmd.exe
 688 Crypserv.exe          0x00010000 FP_NO_HOST_CHECK              NO
 688 Crypserv.exe          0x00010000 NUMBER_OF_PROCESSORS          1
 688 Crypserv.exe          0x00010000 OS                            Windows_NT
 688 Crypserv.exe          0x00010000 Path                          C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\P
Files\GTK\2.0\bin;C:\Program Files\GTK\2.0\bin
 688 Crypserv.exe          0x00010000 PATHEXT                       .COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH
 688 Crypserv.exe          0x00010000 PROCESSOR_ARCHITECTURE        x86
 688 Crypserv.exe          0x00010000 PROCESSOR_IDENTIFIER          x86 Family 6 Model 8 Stepping 1, GenuineIntel
 688 Crypserv.exe          0x00010000 PROCESSOR_LEVEL               6
 688 Crypserv.exe          0x00010000 PROCESSOR_REVISION            0801
 688 Crypserv.exe          0x00010000 ProgramFiles                  C:\Program Files
 688 Crypserv.exe          0x00010000 SystemDrive                   C:
 688 Crypserv.exe          0x00010000 SystemRoot                    C:\WINDOWS
 688 Crypserv.exe          0x00010000 TEMP                          C:\WINDOWS\TEMP
 688 Crypserv.exe          0x00010000 TMP                           C:\WINDOWS\TEMP
 688 Crypserv.exe          0x00010000 USERPROFILE                   C:\Documents and Settings\LocalService
 688 Crypserv.exe          0x00010000 windir                        C:\WINDOWS
 864 DefWatch.exe          0x00010000 ALLUSERSPROFILE               C:\Documents and Settings\All Users
 864 DefWatch.exe          0x00010000 CommonProgramFiles            C:\Program Files\Common Files
 864 DefWatch.exe          0x00010000 COMPUTERNAME                  SPLATITUDE
 864 DefWatch.exe          0x00010000 ComSpec                       C:\WINDOWS\system32\cmd.exe
 864 DefWatch.exe          0x00010000 FP_NO_HOST_CHECK              NO
 864 DefWatch.exe          0x00010000 NUMBER_OF_PROCESSORS          1
 864 DefWatch.exe          0x00010000 OS                            Windows_NT
 864 DefWatch.exe          0x00010000 Path                          C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\P
Files\GTK\2.0\bin;C:\Program Files\GTK\2.0\bin
 864 DefWatch.exe          0x00010000 PATHEXT                       .COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH
 864 DefWatch.exe          0x00010000 PROCESSOR_ARCHITECTURE        x86
 864 DefWatch.exe          0x00010000 PROCESSOR_IDENTIFIER          x86 Family 6 Model 8 Stepping 1, GenuineIntel
 864 DefWatch.exe          0x00010000 PROCESSOR_LEVEL               6
 864 DefWatch.exe          0x00010000 PROCESSOR_REVISION            0801
 864 DefWatch.exe          0x00010000 ProgramFiles                  C:\Program Files
 864 DefWatch.exe          0x00010000 SystemDrive                   C:
 864 DefWatch.exe          0x00010000 SystemRoot                    C:\WINDOWS
 864 DefWatch.exe          0x00010000 TEMP                          C:\WINDOWS\TEMP
 864 DefWatch.exe          0x00010000 TMP                           C:\WINDOWS\TEMP
 864 DefWatch.exe          0x00010000 USERPROFILE                   C:\Documents and Settings\LocalService
 864 DefWatch.exe          0x00010000 windir                        C:\WINDOWS
1076 msdtc.exe             0x00010000 ALLUSERSPROFILE               C:\Documents and Settings\All Users
1076 msdtc.exe             0x00010000 CommonProgramFiles            C:\Program Files\Common Files
1076 msdtc.exe             0x00010000 COMPUTERNAME                  SPLATITUDE
```

It appears using the **consoles** command, but I was unsure what SPLATITUDE meant before Sarah.